

1. Administrative & Study Identification

Full Study Title: Study of Efficacy and Safety of JDQ443 Single-agent as First-line Treatment for Patients With Locally Advanced or Metastatic KRAS G12C- Mutated Non-small Cell Lung Cancer With a PD-L1 Expression < 1% or a PD-L1 Expression ≥ 1% and an STK11 Co-mutation. **Short Title/Acronym:** KontrAST-06 **Protocol Number:** CJDQ443B12201 **NCT Number:** NCT05445843 **Sponsor Name:** Novartis Pharmaceuticals (Novartis Pharma AG) **Investigational Product:** JDQ443 (opnurasib) **Phase of Development:** Phase II **Medical Monitor:** Esra Esencan, Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the antitumor activity and safety of JDQ443 single-agent as first-line treatment for participants with locally advanced or metastatic non-small cell lung cancer (NSCLC) whose tumors harbor a KRAS G12C mutation. **Secondary Objectives:** To evaluate Overall Response Rate (ORR) in Cohort B, Duration of Response (DOR) in both cohorts, Progression-Free Survival (PFS), Overall Survival (OS), safety, pharmacokinetics (PK), and patient-reported outcomes.

2.2 Background & Rationale To address the unmet medical need in patients with KRAS G12C-mutated NSCLC who possess indicators of poor response to standard first-line immunotherapy. Patients whose tumors express low PD-L1 levels (< 1%) or have an STK11 co-mutation typically derive less benefit from immunotherapy. By acting specifically on the KRAS G12C mutation, JDQ443 provides a more personalized, targeted treatment approach aimed at stopping the uncontrolled growth of cancer cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (featuring 2 non-comparative cohorts recruited in parallel) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 120 evaluable patients (Cohort A: \$n=90\$, Cohort B: \$n=30\$) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 30-Jun-2022 **Estimated Primary Completion:** 30-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with locally advanced or metastatic NSCLC. 2. Tumors harbor a KRAS G12C mutation with at least one measurable lesion. 3. **Cohort A:** PD-L1 expression < 1%, regardless of STK11 mutation status. **Cohort B:** PD-L1 expression $\geq$ 1% and an STK11 co-mutation. 4. No prior systemic treatment for metastatic NSCLC.

4.2 Exclusion Criteria 1. Tumors harboring an EGFR-sensitizing mutation and/or ALK rearrangement. 2. Previous use of a KRAS G12C inhibitor. 3. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 4. Concurrent use of prohibited medications, such as strong CYP3A inducers, that cannot be discontinued at least 7 days prior to the first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: JDQ443 200 mg administered twice daily (BID). **Route of Administration:** Oral. **Duration of Treatment:** Administered in 21-day cycles until disease progression, unacceptable toxicity, or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing underlying comorbidities. **Prohibited:** Strong CYP3A inducers (must be washed out at least 7 days prior to treatment) and other systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Biomarker confirmation (KRAS G12C, PD-L1, STK11), Baseline Imaging
Baseline	Cycle 1 Day 1	Enrollment, First Dose of JDQ443, Vital Signs, Safety Labs
Interim Visits	Day 1 of each 21-day Cycle	JDQ443 Dispensation, Safety Labs, PK sampling, AE Assessments
Tumor Assessment	Every 6-8 Weeks	Imaging for objective response evaluation per RECIST 1.1
End of Study	+30 Days post-last dose	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Response Rate (ORR), defined as the percentage of participants with a confirmed complete response (CR) or partial response (PR) as their best overall response in Cohort A. **Measurement Method:** Objective

radiological assessment evaluated by a Blinded Independent Review Committee (BIRC) per RECIST 1.1 criteria.

7.2 Secondary & Exploratory Endpoints 1. Overall Response Rate (ORR) in Cohort B. 2. Duration of Response (DOR) in both cohorts. 3. Progression-Free Survival (PFS) and Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient interviews, physical examinations, and routine laboratory monitoring throughout the 21-day cycles. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review committee evaluates the ongoing safety/tolerability of JDQ443 dosing.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy analysis is based on the Intent-to-Treat (ITT) set of enrolled patients with measurable disease; the Safety Set includes all patients receiving at least one dose of JDQ443. **Sample Size Justification:** Approximately 120 patients (90 in Cohort A; 30 in Cohort B). Designed to demonstrate clinically relevant efficacy in Cohort B if an ORR of $\geq 40\%$ is observed, with the lower bound of the 95% confidence interval $\geq 20\%$. **Handling of Missing Data:** Patients without valid post-baseline tumor assessments are treated as non-responders in the ORR calculation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote source data verification by clinical research associates (CRAs). **Audit Trail:** Standardized eCRF data entry, central blinded imaging review (BIRC), and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: CJDQ443B12201 **Registry Number:** NCT05445843 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** KonTRASt-06 **Official Regulatory Title:** Study of Efficacy and Safety of JDQ443 Single-agent as First-line Treatment for Patients With Locally Advanced or Metastatic KRAS G12C- Mutated Non-small Cell Lung Cancer With a PD-L1 Expression $< 1\%$ or a PD-L1 Expression $\geq 1\%$ and an STK11 Co-mutation.

12. Clinical Framework (JDQ443 Specific)

Primary Condition: Locally Advanced or Metastatic Non-small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Tumor harbors KRAS G12C mutation; Cohort A requires PD-L1 < 1%; Cohort B requires PD-L1 $\geq 1\%$ and STK11 co-mutation. **Medical Methodology:** Targeted therapy (KRAS G12C inhibitor) as first-line treatment. **Optimization Target:** Objective tumor shrinkage and durable disease control.

13. Study Procedures & Methodology

Management Pathway: Targeted continuous oral monotherapy. **Education Protocol (HCP):** Investigator training on recognizing and managing potential AEs associated with JDQ443. **Education Protocol (Patient):** Strict dosing adherence training for daily oral administration (200 mg BID) and immediate AE reporting. **Quality Audit Mechanism:** Blinded Independent Review Committee (BIRC) ensuring unbiased radiological evaluation per RECIST 1.1.

14. Observation & Follow-Up Schedule

Total Duration: Continuous in 21-day cycles until disease progression or unacceptable toxicity. **Onsite Visit Cadence:** Day 1 of each 21-day cycle. **Remote Monitoring Frequency:** Intermittent per sponsor monitoring plan. **Checklist Review Interval:** Radiological tumor assessments performed every 6 to 8 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	Esra Esencan	_____	[DD-MMM-YYYY]				
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]				